

Cognitive Robotics – 00970244

Autonomous Multi-Strategy Robot System for Rapid Person Detection in Unknown Indoor Environments

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Simulation video

<https://youtu.be/mpGusTGO4eU>



Project Idea

Development of an autonomous mobile robot system (TurtleBot3) that explores unknown indoor environments to detect and log human presence using computer vision and LiDAR sensor fusion. The robot operates without ROS middleware, implementing a standalone controller that combines:

- Real-time person detection using YOLOv8
- Autonomous exploration with multiple navigation strategies
- Sensor fusion (Camera + LiDAR + GPS + IMU)
- Obstacle avoidance and recovery behaviors

Primary Goal

Create a robust search and rescue simulation where the robot autonomously explores an environment, detects all people present, and records their positions without human intervention

Implementation Environment

Primary Platform: Webots R2025a

- Physics engine: ODE (Open Dynamics Engine)
- Real-time simulation with 20ms timestep
- TurtleBot3 Burger model with:
 - Differential drive (2 wheels)
 - 360° LiDAR (LDS-01)
 - RGB Camera
 - GPS sensor
 - IMU (Inertial Measurement Unit)

Indoor Environment:

- 50m × 50m office/apartment space
- Walls, furniture, obstacles
- 5-6 pedestrian models distributed throughout
- Dynamic lighting conditions

Technology Stack:

- Python 3.x
- Webots Controller API
- OpenCV (image processing)

- Ultralytics YOLOv8n (person detection)
- NumPy (numerical computations)

Problem Definition & Planning Algorithms

Problem Type:

Multi-Objective Exploration with Object Detection

- Exploration: Unknown environment coverage
- Detection: Real-time person identification
- Localization: World coordinate mapping
- Constraint: No prior map, reactive navigation only

Navigation Algorithms Implemented:

1. Reactive Navigation (Bug Algorithm Inspired)
 - Pure sensor-based obstacle avoidance
 - No path planning overhead
 - Real-time response to sensor data
 - Reference: [Bug Algorithms for Robot Navigation](#)
2. Wall Following (Trémaux's Algorithm Concept)
 - Right-hand rule for systematic coverage
 - Maintains constant wall distance (0.4m)
 - Reference: [Wall Following Algorithms](#)
3. Finite State Machine (FSM) Control
 - Four navigation states with automatic transitions
 - Stuck detection and recovery behaviors
 - Reference: [FSM for Robot Navigation](#)
4. Sensor Fusion Architecture
 - Camera + LiDAR data fusion for person localization
 - Kalman-like filtering of duplicate detections
 - Reference: [Multi-Sensor Fusion](#)

Key Papers/Resources:

- YOLOv8: [Ultralytics YOLO](#)
- Differential Drive Kinematics: [Mobile Robot Kinematics](#)

- Bug Algorithms: Choset et al., "Principles of Robot Motion"

Performance Metrics

Primary Metrics:

1. Exploration Efficiency

- Area covered per time unit (m^2/minute)
- Total exploration time to complete environment
- Path length vs. environment size ratio

2. Detection Accuracy

- Person detection rate (True Positives / Total Persons)
- False positive rate
- Detection precision (confidence threshold analysis)
- Position accuracy (error in meters from ground truth)

3. Navigation Robustness

- Number of stuck incidents
- Average recovery time
- Collision frequency
- Smooth navigation score (acceleration variance)
- CPU usage (%)
- Memory consumption
- YOLO inference time (ms)
- Control loop frequency (Hz)

Simulation Scenarios

Environment Complexity Variation

- Simple: Open space, few obstacles (2-3 walls)
- Medium: Office layout with furniture
- Complex: Maze-like apartment with narrow corridors
- Metric Focus: Exploration time, stuck incidents

Person Density

- Low Density: 2-3 people in 50m² area
- Medium Density: 5-7 people
- High Density: 10+ people in clusters
- Metric Focus: Detection rate, false positives
- Scenario 3: Lighting Conditions
- Bright: Full office lighting
- Dim: Reduced lighting (shadows)
- Dynamic: Moving light sources
- Metric Focus: YOLO confidence scores

Speed vs. Accuracy Trade-off

- Conservative: 0.3 m/s, YOLO every 3 steps
- Balanced: 0.5 m/s, YOLO every 5 steps
- Aggressive: 0.8 m/s, YOLO every 7 steps
- Metric Focus: Coverage speed vs. detection accuracy

Robot Configuration

- Single Robot: Full autonomous operation
- Multi-Robot: 2-3 robots (future extension)
- Different Sensors: Camera-only vs. LiDAR+Camera
- Metric Focus: System scalability

3 Major Improvements

Adaptive Multi-Robot Coordination System

Instead of using a single robot to explore the entire environment, deploy 2-4 TurtleBot3 robots simultaneously that work together as a coordinated team. Each robot explores a different section of the environment while sharing information about detected people in real-time. This dramatically reduces total exploration time through parallel processing and distributed workload.

Dynamic Person Tracking with Re-Identification

Instead of treating each person detection as an independent event, continuously track people over time as they move through the environment. The system assigns a unique ID to each person, follows their trajectory, predicts where they'll go next, and can re-identify them even if they temporarily leave the camera view.

Semantic SLAM with Priority-Based Navigation

Build a rich, semantic map of the environment that doesn't just show "obstacle here" but understands "This is a chair, that's a door, there's a person lying down." Use this understanding to make intelligent navigation decisions, especially prioritizing critical situations (e.g., person lying motionless = highest urgency).

Current system sees world as: occupied/free space This improvement sees world as: labeled objects with meaning